

Firecracker microVM performance: AMD EPYC on OCI and AWS

OCI bare-metal · Ashburn (us-ashburn-1) · Firecracker v1.16.1 · kernel 6.8.0-1057-oracle · THP madvise · generated from run 2026-08-28T22:19:15Z

Note on Intel: OCI has no same-generation Intel bare-metal shape (the only Intel bare-metal, [BM.Standard4.Ax](#), was not available in-region), so this study compares AMD generations — EPYC **Genoa (Zen4)** vs **Turin (Zen5)** — rather than AMD vs Intel. A companion AWS study (AMD Turin vs Intel Granite Rapids) remains the cross-vendor reference.

Executive summary

The same five-metric Firecracker suite was run on three OCI AMD bare-metal shapes under an identical Firecracker build, guest kernel, and golden rootfs. The core-matched pair — **E5.192 (Genoa/Zen4)** and **E6.Ax.192 (Turin/Zen5)**, both 384 threads — isolates the generational step; **E6.256** is the larger 512-thread flagship.

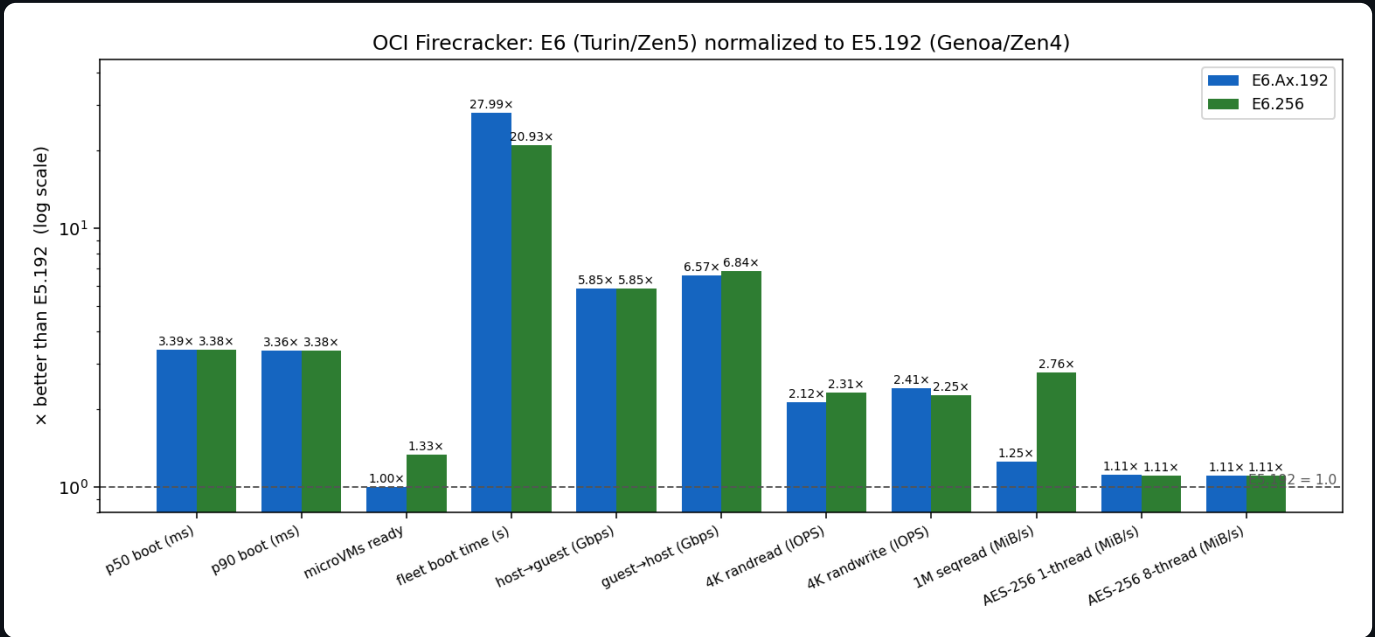
The cleanest, storage-independent generational signal is **in-guest AES-256** (AES-NI/VAES): per-thread it rises from **980.8** → **1,089.6 MiB/s** (1.11× higher (**E6.Ax**)), and at 8 threads **7,845.9** → **8,676.2 MiB/s** (1.11× higher (**E6.Ax**)) — a ~8–11% per-thread Zen5 uplift.

The other metrics are reported **as-shipped**: E6 shapes include **local NVMe**, while **E5.192 has none** and ran on a standard iSCSI block volume (the only option for that shape). Boot latency, density fleet-boot, and block IOPS therefore reflect the **shape as a package** (CPU *and* its bundled storage), not the CPU alone — which is exactly what a customer deploying the shape experiences. See the storage note below.

Full results

Metric	E5.192 (Zen4)	E6.Ax.192 (Zen5)	E6.256 (Zen5)
Boot latency (cold-start, fast-init)			
p50 boot (ms) †	414.1	122.3	122.5
p90 boot (ms) †	415.9	123.7	123.2
Fleet density (256 MiB/VM)			
microVMs ready	384	384	512
fleet boot time (s) †	48.98	1.75	2.34
Network (virtio-net, iperf3)			
host→guest (Gbps)	8.7	50.8	50.7
guest→host (Gbps)	8.6	56.4	58.7
Block I/O (virtio-blk, fio direct)			
4K randread (IOPS) †	110,256	233,852	255,019
4K randwrite (IOPS) †	88,539	213,707	199,577
1M seqread (MiB/s) †	8,507	10,652	23,476
Guest compute (in-VM)			
AES-256 1-thread (MiB/s)	980.8	1,089.6	1,086.7
AES-256 8-thread (MiB/s)	7,845.9	8,676.2	8,681.6

† = storage-sensitive (E6 on local NVMe, E5 on iSCSI block volume).



Each metric normalized to E5.192 = 1.0 (dashed line); bars show how many times better E6.Ax.192 and E6.256 are (for boot latency and fleet-boot, "better" means lower, so the ratio is inverted). Log scale — the improvements span 1.1x (AES) to ~28x (fleet boot), so a linear axis would hide the smaller ones. Storage-fed metrics (boot, block, fleet-boot) carry E6's local-NVMe advantage; AES is the pure CPU-generation step.

Methodology notes

Storage is a real shape difference, not normalized away. E6.Ax.192 and E6.256 ship with local NVMe; E5.192 has no local NVMe, so its Firecracker working set (guest rootfs + overlays) ran on an attached iSCSI block volume. We deliberately did **not** normalize storage (e.g. via a RAM disk) because a customer deploys the shape as-shipped: an E6 tenant gets local NVMe, an E5 tenant must attach block storage. The storage-sensitive rows (†) therefore reflect the true as-deployed experience. The pure CPU-generation signal is carried by AES-256, which is storage-independent.

Density is thread-capped. At 256 MiB/VM every host saturates at one microVM per hardware thread (E5.192 & E6.Ax.192: 384; E6.256: 512) long before RAM is exhausted, so density scales with thread count, not a memory advantage.

Network gap larger than the CPU step alone. The E5→E6 virtio-net ratio is much bigger than the ~1.1x seen on the pure-CPU AES metric. virtio-net throughput is driven by the host vhost worker and is sensitive to NUMA placement of the guest vCPUs, the TAP, and the vhost thread — none of which the harness pins. On the dual-socket E5 this can split the data path across sockets and depress throughput. The E5 network figure (measured consistently at ~8.6 Gbps across directions) should be treated as a floor pending a NUMA-pinned re-measurement, not as the ceiling of Genoa's virtio-net capability.

matrixprod excluded. A stress-ng matrixprod stressor is collected but excluded from conclusions (it dispatches into Intel AMX on Intel parts — an ISA AMD does not expose). AES-256-GCM is the retained compute benchmark; both AMD generations support AES-NI/VAES.

Test configuration

SHAPE	CPU	SOCKETS × CORES	THREADS	RAM	LOCAL STORAGE
E5.192 (Genoa / Zen4)	AMD EPYC 9J14 96-Core Processor	2 × 96	384	2267 GiB	none (iSCSI block volume)
E6.Ax.192 (Turin / Zen5)	AMD EPYC 9J45 128-Core Processor	2 × 96	384	1511 GiB	2×960 GB NVMe
E6.256 (Turin / Zen5)	AMD EPYC 9J45 128-Core Processor	2 × 128	512	3023 GiB	2×960 GB NVMe

OCI cost & price-performance

List prices (US\$, pre-discount): OCI E-series bills **\$0.0250 per OCPU-hour + \$0.0015 per GB-hour**, and **E6 is priced the same per-OCPU as E5** — so E6's performance gains come at no per-core premium. Bare-metal OCPU/memory are fixed; 1 OCPU = 2 vCPUs (2 hardware threads). These are list rates before any committed-use or negotiated discount — override with `OCI_OCPU_RATE` / `OCI_MEM_RATE` to use your own.

SHAPE	OCPU	RAM (GB)	OCPU \$/HR	MEMORY \$/HR	TOTAL \$/HR	~\$/MONTH (730 H)
E5.192	192	2,267	4.80	3.40	8.20	5,987

SHAPE	OCPU	RAM (GB)	OCPU \$/HR	MEMORY \$/HR	TOTAL \$/HR	~\$/MONTH (730 H)
E6.Ax.192	192	1,511	4.80	2.27	7.07	5,159
E6.256	256	3,023	6.40	4.53	10.93	7,982

Price-performance: microVM density economics

For a Firecracker / serverless host the money metric is **cost per microVM-hour** — how cheaply you can host each 256-MiB guest — and its inverse, microVMs per dollar-hour. (microVM count is thread-capped, so it scales with OCPU; the differentiator is how much you pay in memory for those threads.)

SHAPE	MICROVMS (256 MiB)	TOTAL \$/HR	\$ PER MICROVM-HOUR	MICROVMS PER \$/HR
E5.192	384	8.20	\$0.0214	46.8
E6.Ax.192	384	7.07	\$0.0184	54.3
E6.256	512	10.93	\$0.0214	46.8

Cost per microVM-hour in US\$; lower is better. Density is one microVM per hardware thread at 256 MiB.

Read: on pure density economics, **E6.Ax.192** is the best value at **\$0.0184 per microVM-hour** — it packs the most guests per dollar. E6.256 costs more per hour (more OCPUs + memory) but also hosts more microVMs, so its per-microVM cost lands close to E5; the E6 win is that you get Turin's ~2× throughput and local NVMe at the **same per-OCPU price** as E5. Because the storage-fed metrics (boot, block) are far better on E6's local NVMe, E6's *performance* per dollar on those axes is dramatically higher than the raw density numbers alone suggest.

Pricing is OCI public list price and may not reflect your committed-use/negotiated rate; treat the dollar figures as relative, not billing-accurate.

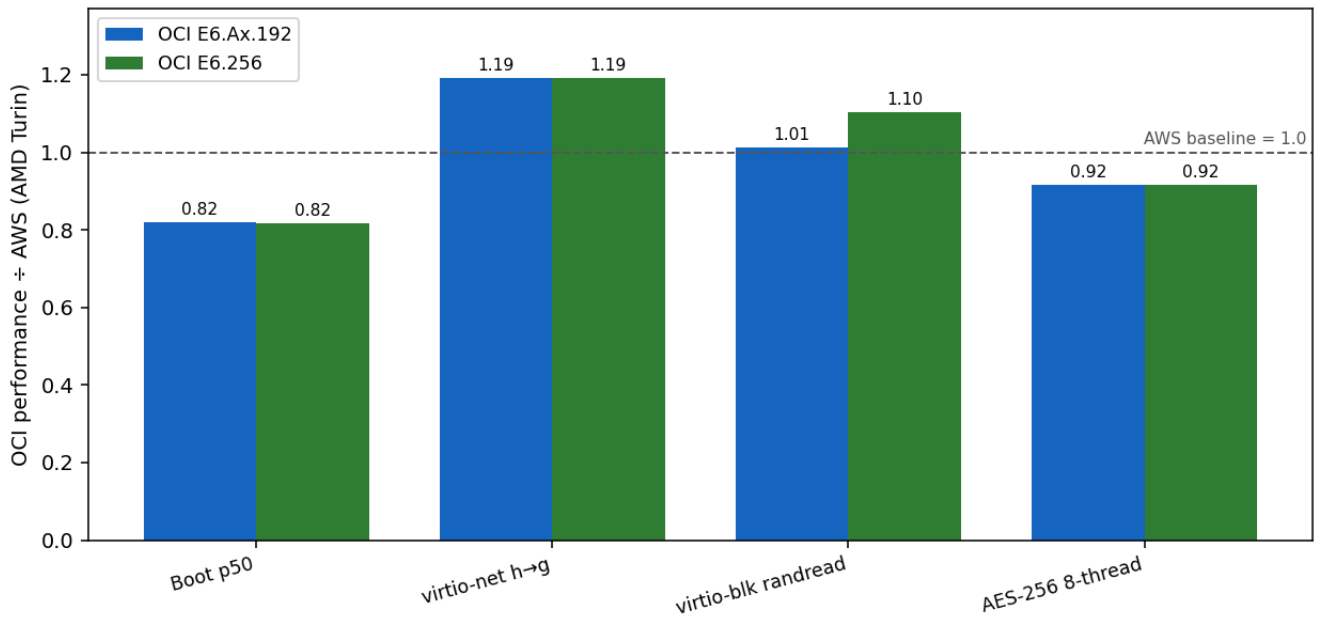
Cross-cloud: AMD Turin on OCI vs AWS

Same silicon generation (AMD EPYC Turin/Zen5), different cloud. OCI **E6.256** (local NVMe) vs the two AWS Turin metal hosts from the companion AWS study — **c8a** (compute) and **m8a** (general-purpose), both EPYC 9R45, Turin + local NVMe. The two AWS hosts share the same silicon (they differ only in RAM class), so their per-guest numbers are near-identical; the ratio column compares OCI to their geometric mean (the same pair-geomean method used in the AWS study). This isolates the *cloud platform* (host kernel, VMM tuning, storage, power limits) on matched silicon.

METRIC	C8A (AWS)	M8A (AWS)	OCI E6.AX.192	OCI E6.256	E6.AX VS AWS	E6.256 VS AWS
Boot p50 (ms)	99.7	100.5	122.3	122.5	0.82× (AWS)	0.82× (AWS)
virtio-net h→g (Gbps)	45.4	40.0	50.8	50.7	1.19× (OCI)	1.19× (OCI)
virtio-blk randread (IOPS)	209,482.0	254,866.0	233,852.0	255,019.0	1.01× (OCI)	1.10× (OCI)
AES-256 8-thread (MiB/s)	9,482.8	9,478.8	8,676.2	8,681.6	0.92× (AWS)	0.92× (AWS)

Ratios normalized so >1.0 means OCI beats the AWS AMD baseline on that metric (boot is inverted — lower latency is better). Baseline = geomean of c8a and m8a. The two OCI Turin shapes track each other closely, as expected for identical silicon.

Firecracker on AMD Turin: OCI normalized to AWS baseline



Bars: OCI E6.Ax.192 and E6.256 normalized to the AWS AMD Turin baseline (dashed line = AWS = 1.0). Above the line = OCI advantage.

Read: OCI's AMD Turin leads on virtio-net and virtio-block but trails slightly on cold-start boot and full-socket AES — a mixed result, not a clean sweep. The boot and AES gaps are small and plausibly reflect host-kernel/VMM tuning and per-SKU clock/power-limit differences (OCI 9J45 vs AWS 9R45 custom Turin parts), not an architectural difference.